

Since μ'_i and μ_i are orthogonal to $\tilde{X}'_i$ and $\tilde{X}_i$, and the constraint $\text{Cov}(X', Z) = \mathbf{0}$ is invariant to constant shifts, we can optimize the two terms in Eq. 7 independently. The first term is trivial: it is minimized when $\mu'_i = \mu_i$, and hence $X'_i = \tilde{X}'_i + \mathbb{E}[X_i]$.

Orthogonal projection. We can now rewrite the zero covariance condition as an orthogonality constraint on $\tilde{X}_i$. Specifically, for every $i \in \{1 \dots d\}$ we have

$$\underset{\tilde{X}'_i \in \mathcal{H}}{\text{argmin}} \|\tilde{X}'_i - \tilde{X}_i\|_{\mathcal{H}}^2 \quad \text{s.t.} \quad \forall j \in \{1 \dots k\} : \langle \tilde{X}'_i, \tilde{Z}_j \rangle_{\mathcal{H}} = 0, \quad (8)$$

where $\tilde{Z} = Z - \mathbb{E}[Z]$. In other words, we seek the nearest $\tilde{X}'_i$ to $\tilde{X}_i$ orthogonal to $\mathcal{Z} = \text{span}(\{\tilde{Z}_1, \dots, \tilde{Z}_k\})$, which is simply the orthogonal projection of $\tilde{X}_i$ onto $\mathcal{Z}^\perp$. This in turn is equal to the ordinary least squares residual from regressing $\tilde{X}$ on $\tilde{Z}$:

$$\tilde{X}'_i = \tilde{X}_i - \text{proj}(\tilde{X}_i, \mathcal{Z}) = X_i - (\Sigma_{XZ})_i \Sigma_{ZZ}^+(Z - \mathbb{E}[Z]) - \mathbb{E}[X_i]. \quad (9)$$

Putting it all together. Plugging Eq. 9 into $X'_i = \tilde{X}'_i + \mathbb{E}[X_i]$ and combining all components into vector form yields

$$X'_{\text{LEACE}} = X - \Sigma_{XZ} \Sigma_{ZZ}^+(Z - \mathbb{E}[Z]), \quad (10)$$

which completes the proof. $\square$

J Notation Key

$\mathcal{Z}$	The space of one-hot labels $\{(z_1, \dots, z_k) \in \{0, 1\}^k \mid \sum_{j=1}^k z_j = 1\}$ (treated interchangeably with the integers $\{1, \dots, k\}$ when convenient).
X, Z	Integrable (i.e. finite first moment) random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively (or their realized values inside an expectation, e.g. in $\mathbb{E}[f(X)]$). Z is sometimes restricted to the one-hot labels $\mathcal{Z}$, in which case we assume each $\mathbb{P}(Z = j) > 0$.
X_i, Z_j	The i^{th} and j^{th} components thereof, themselves scalar random variables (or their realized values inside an expectation).
ξ, ζ	Scalar random variables taking values in $\mathbb{R}$.
η	A predictor function $\mathbb{R}^d \rightarrow \mathcal{Z}$ (or its value $\eta(X)$ when inside an expectation).
$\mathcal{V}$	A space of predictor functions $\{\eta(\cdot; \theta) : \mathbb{R}^d \rightarrow \mathbb{R}^k \mid \theta \in \Theta\}$, parameterized by θ and containing all constant functions.
$\mathcal{L}$	A space of loss functions $\{\mathcal{L} : \mathbb{R}^k \times \mathcal{Z} \rightarrow [0, \infty)\}$.
r	An erasure function $\mathbb{R}^d \rightarrow \mathbb{R}^d$, hopefully making a minimal edit to X that eliminates the ability to predict labels Z with predictors in $\mathcal{V}$.
$\mathbf{A}$	A matrix with entries in $\mathbb{R}$.
A_{ij}	The entry thereof at the i^{th} row and j^{th} column.
$\mathbf{A}^+$	The Moore-Penrose pseudoinverse of $\mathbf{A}$.
$\mathbf{v}$	A column vector with entries in $\mathbb{R}$.
v_i	The i^{th} component thereof.

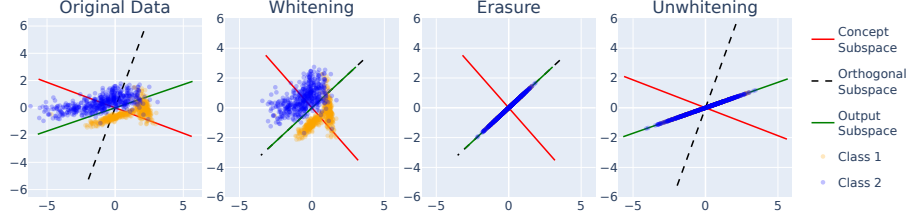


Figure 1: LEACE projection on a toy dataset in 3 steps. First the data is whitened using the linear transformation $W = (\Sigma_{XX})^{-1/2}$, ensuring equal variance in all directions. It is then orthogonally projected onto $\text{colsp}(W\Sigma_{XZ})^\perp$, guaranteeing linear guardedness. Finally, we unwhiten the data with the pseudoinverse W^+ so that its covariance structure mimics the original.

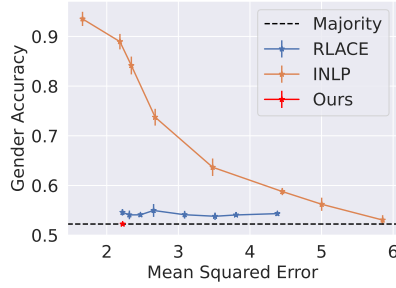


Figure 2: Gender prediction accuracy after debiasing against the mean squared distance of the debiased embedding from the original. Ours is the only method to achieve *perfect* erasure on this dataset (dashed line), and does so with low MSE. For INLP and RLACE, each point corresponds to a different setting of the rank hyperparameter; from left to right the ranks are 1, 4, 8, 16, 32, 50, 64, and 100. Our method is rank 1.

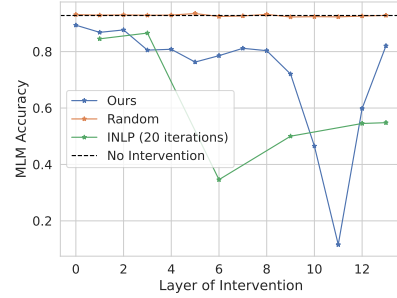


Figure 3: Corrected version of Figure 3 in our submission, which had erroneously reported that BERT exhibited *better* masked language modeling accuracy after a random subspace of its activations was ablated. One of our authors had accidentally transposed a digit in the “No Intervention” MLM accuracy, typing 0.892 in lieu of 0.928.